

CASTLEDERG, United Kingdom

WMO: 039040

Lat: 54.7071N

Lon: 7.5775W

Elev: 50

StdP: 100.73

Time Zone: 0.00 (EUW)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -5.3	(c) -3.1	(d) -6.4	(e) 2.2	(f) -4.3	(g) -4.3	(h) 2.6	(i) -1.6	(j) 15.0	(k) 8.6	(l) 13.4	(m) 8.9	(n) 0.8	(o) 290	(p) 0.614

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.8	22.7	17.5	20.9	16.5	19.4	15.7	18.4	21.4	17.3	19.7	16.5	18.6	3.7	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.1	12.3	19.6	16.3	11.6	18.5	15.5	11.0	17.6	52.0	21.4	48.9	19.8	46.3	18.7	25.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 11.3	(b) 9.7	(c) 8.5	(e) -7.6	(f) 26.9	(g) 3.4	(h) 2.0	(i) -10.1	(j) 28.4	(k) -12.0	(l) 29.6	(m) -13.9	(n) 30.7	(o) -16.4	(p) 32.2		
(a) 11.3	(b) 9.7	(c) 8.5	(e) -7.7	(f) 20.9	(g) 3.2	(h) 1.8	(i) -10.1	(j) 22.2	(k) -11.9	(l) 23.3	(m) -13.8	(n) 24.3	(o) -16.1	(p) 25.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.2	4.6	5.1	6.1	7.9	10.6	13.3	14.7	14.3	12.6	9.7	6.7	4.8		
	DBStd	4.60	3.21	2.91	2.73	2.63	2.67	2.47	2.27	1.93	2.42	2.79	2.94	3.90		
	HDD10.0	842	167	139	124	75	23	2	0	0	4	39	104	163		
	HDD18.3	3331	424	371	380	314	238	153	116	127	172	267	349	419		
	CDD10.0	558	1	2	3	11	43	101	146	132	83	31	5	2		
	CDD18.3	4	0	0	0	0	0	1	3	1	0	0	0	0		
	CDH23.3	46	0	0	0	0	2	14	27	1	2	0	0	0		
	CDH26.7	4	0	0	0	0	0	2	2	0	0	0	0	0		
	WSAvg	3.5	4.2	4.1	4.0	3.6	3.5	3.2	3.0	3.1	3.2	3.4	3.5	3.9		
	PrecAvg	1212	130	101	82	75	86	79	85	95	91	121	132	136		
(15) Precipitation	PrecMax	1574	208	205	168	123	150	143	149	168	173	229	248	281		
	PrecMin	915	19	34	25	31	11	35	36	21	7	31	50	44		
	PrecStd	175	53	50	32	26	37	30	34	33	44	57	47	64		
	PrecStd	175	53	50	32	26	37	30	34	33	44	57	47	64		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.7	13.0	15.5	19.0	23.4	25.0	26.5	22.3	21.6	17.2	14.6	13.3		
		MCWB	11.3	11.3	11.1	13.2	17.3	17.6	20.1	17.5	17.4	15.3	13.1	12.0		
	2%	DB	11.4	11.7	13.0	16.3	20.3	22.2	23.0	20.3	19.4	15.8	13.1	12.1		
		MCWB	10.4	10.4	10.3	11.9	15.3	16.8	18.2	16.6	16.0	14.1	12.2	11.0		
	5%	DB	10.5	10.7	11.6	14.5	17.9	20.1	20.8	18.9	18.0	14.9	12.2	11.1		
		MCWB	9.6	9.6	9.5	11.1	13.7	15.9	16.8	15.8	15.5	13.5	11.2	10.2		
	10%	DB	9.5	9.7	10.6	12.9	15.9	18.1	18.9	17.9	16.7	14.0	11.3	10.1		
		MCWB	8.6	8.5	8.8	10.2	12.3	14.7	15.7	15.2	14.5	12.6	10.2	9.2		
	0.4%	WB	11.7	11.7	12.0	14.3	17.9	19.0	20.7	18.6	18.2	16.0	13.5	12.3		
		MCDB	12.4	12.5	14.0	17.0	22.1	23.6	25.4	20.9	20.0	16.8	14.1	13.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	10.6	10.7	10.9	12.7	15.8	17.5	18.8	17.5	16.9	14.7	12.3	11.2		
		MCDB	11.3	11.5	12.4	15.2	19.6	21.1	22.3	19.3	18.5	15.4	12.9	11.9		
	5%	WB	9.7	9.8	10.1	11.6	14.3	16.2	17.3	16.6	15.9	13.7	11.4	10.4		
		MCDB	10.5	10.6	11.3	13.9	17.0	19.2	19.9	18.3	17.4	14.7	12.1	11.1		
	10%	WB	8.7	8.8	9.1	10.5	13.0	15.1	16.3	15.8	15.0	12.9	10.4	9.2		
		MCDB	9.5	9.7	10.3	12.6	15.3	17.5	18.4	17.4	16.3	13.8	11.2	10.0		
	5% DB	MDBR	5.3	6.1	7.4	8.6	8.9	8.3	7.8	7.5	7.6	6.9	5.9	5.5		
		MCDBR	5.6	5.8	8.4	11.1	12.5	11.9	10.8	8.8	8.9	6.8	5.7	5.8		
		MCWBR	5.4	5.1	6.1	7.3	8.1	7.2	6.6	5.8	6.0	5.4	5.3	5.6		
		MCWBR	5.6	5.6	7.4	9.7	10.9	10.4	9.8	7.6	7.7	6.1	5.8	5.9		
(35) Mean Daily Temperature Range	5% WB	MCWBR	5.5	5.2	5.8	6.9	7.6	6.8	6.4	5.7	5.9	5.3	5.6	5.7		
		MCWBR	5.5	5.2	5.8	6.9	7.6	6.8	6.4	5.7	5.9	5.3	5.6	5.7		
		MCWBR	5.5	5.2	5.8	6.9	7.6	6.8	6.4	5.7	5.9	5.3	5.6	5.7		
		MCWBR	5.5	5.2	5.8	6.9	7.6	6.8	6.4	5.7	5.9	5.3	5.6	5.7		
(40) Clear-Sky Solar Irradiance	taub		0.344	0.351	0.374	0.384	0.391	0.378	0.380	0.365	0.361	0.348	0.334	0.332		
	taud		2.193	2.271	2.253	2.268	2.291	2.364	2.363	2.421	2.426	2.385	2.261	2.164		
	Ebn at Noon		560	717	790	838	853	869	858	852	804	718	574	480		
	Edh at Noon		64	83	104	118	121	114	112	100	88	73	59	53		
(44) All-Sky Solar Radiation	RadAvg		0.57	1.25	2.29	3.71	4.71	4.80	4.39	3.59	2.58	1.47	0.70	0.42		
	RadStd		0.06	0.11	0.22	0.35	0.42	0.48	0.36	0.30	0.14	0.13	0.07	0.04		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (26 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page